

Beyond Today's Price: Predicting the Direction of Football Player Market Values from Recent Performance

Soccer

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Abstract.

Player market value influences recruitment, retention, and transfer timing, but today's valuation already carries much of the information used to estimate tomorrow's price. For clubs, the more actionable question is whether recent football performance helps distinguish players whose values are more likely to rise, remain stable, or fall. We examine whether that recent performance adds forward-looking information beyond current market value and valuation history.

All features use only information available before each valuation date. We begin with current market value, add six valuation-history features, and then introduce 25 recent-performance features spanning playing time, goalscoring, defending, chance creation, and possession progression. Evaluation uses a locked English Championship test set of 752 valuation events from 606 players, followed by external testing across 27 competitions. A price-only model (M1) is compared with a performance-informed model (M5) using residual MAE, three-class direction accuracy, Spearman rank correlation, shortlist behavior, and material-movement detection. Uncertainty is assessed with 2,000 player-clustered bootstrap draws so repeated observations from the same player remain grouped.

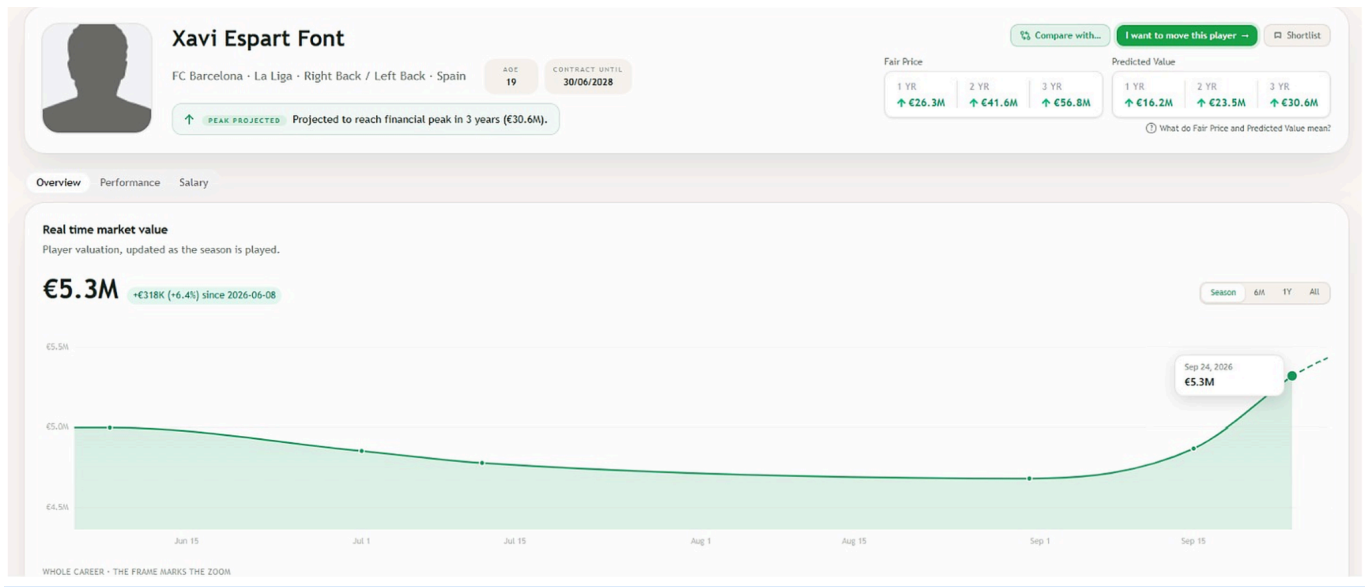


Figure 1. Player valuation dashboard combining current market value with one-, two-, and three-year fair-price and predicted-value estimates, providing a forward-looking view of the player's potential financial trajectory.

On the locked Championship test, M5 improves each tracked point estimate. Residual MAE decreases from 0.3066 to 0.2957, direction accuracy increases from 35.6% to 46.7%, and Spearman correlation rises from 0.278 to 0.296. Direction accuracy also exceeds the 42.3% majority-class benchmark. The clearest gain is therefore directional rather than exact-price prediction. Across all 27 external competitions, M5 improves direction accuracy over M1, while exact-value improvement occurs in only 8 of 27.

Recent performance is not equally informative across feature groups. Playing time contributes the strongest ranking signal, followed by attacking, goalkeeping, defending, creation, and possession progression. Bootstrap intervals support the direction and exact-value differences, but not a reliable difference in ranking quality. Neither model identifies any of the 10 pre-declared extreme breakout rises, so the approach should not be interpreted as a breakout detector or an automated valuation engine.

The practical use is shortlist prioritization. A club can start with its existing scouting list and current valuations, then use recent performance to re-rank players as likely risers, stable players, or fallers before human review. The evidence is strongest once roughly five prior matches are available. The model is therefore best viewed as an additional directional signal for recruitment, retention, and transfer-timing decisions, while exact future prices and exceptional breakouts remain substantially harder to generalize.

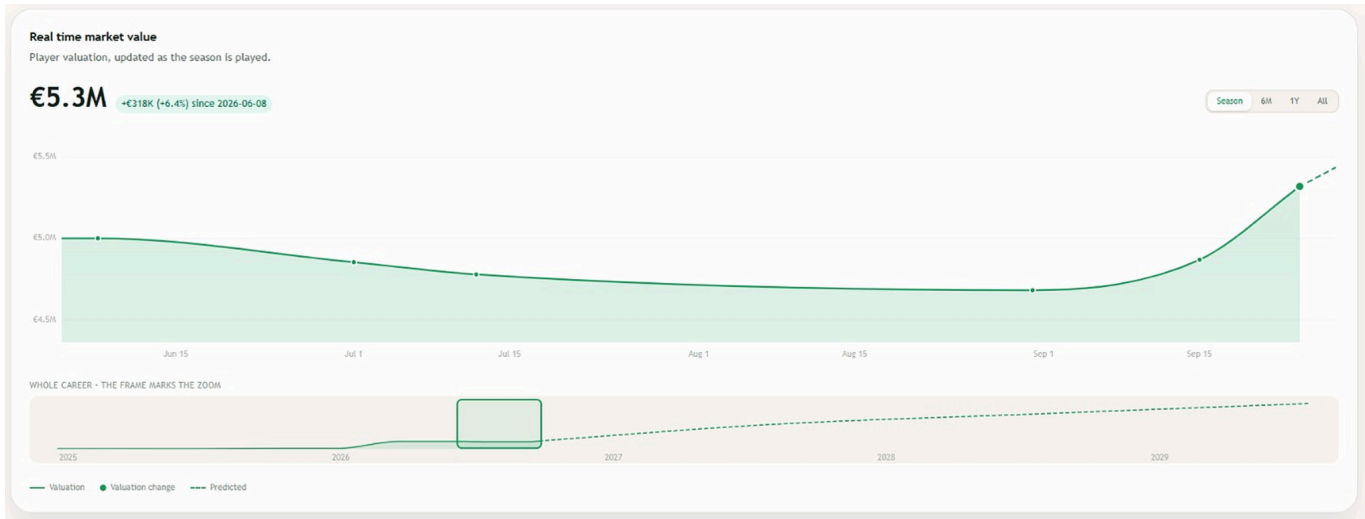


Figure 2. Real-time player market-value trajectory showing observed valuation changes through the season and the projected future path, allowing clubs to assess how a player's value may evolve beyond the current valuation.

This research was developed in collaboration with SoccerSolver. SoccerSolver currently works with more than 10 football clubs.

Open-source repository: [PUBLIC REPOSITORY URL]